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ENGINEERING



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Saveetha Institute of Medical And Technical Sciences
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MORPHOLOGY AND WORD FORMATION IN NATURAL LANGUAGES

DSA0302 – Natural Language Processing

Guided By:

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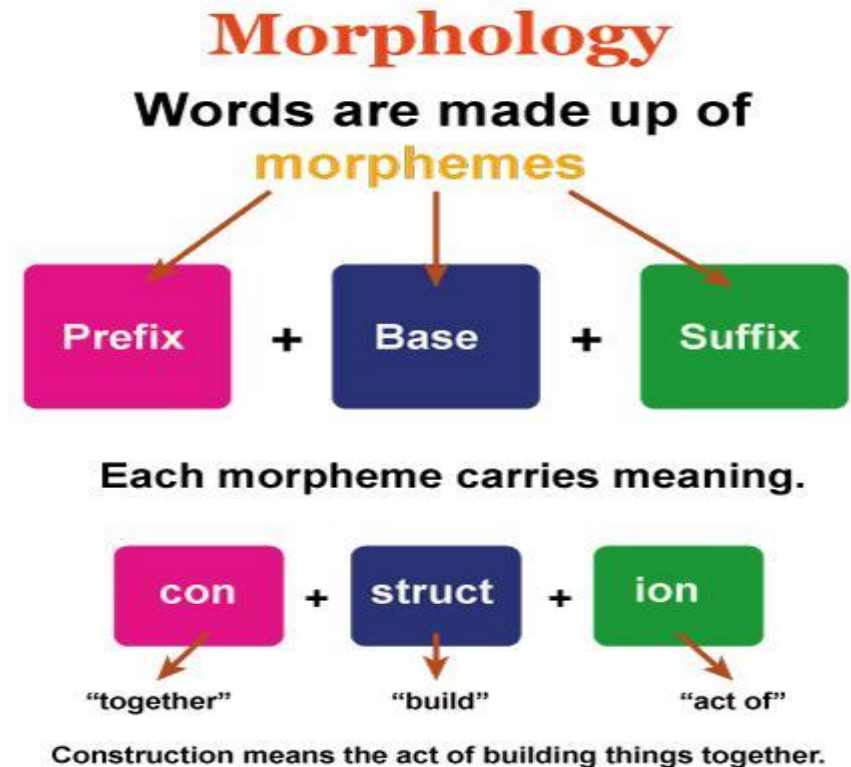
INTRODUCTION



- Morphology is the study of the internal structure of words and how new words are formed using the smallest meaningful units called **morphemes**.

➤ Key Objectives

- Understand word structure.
- Analyze prefixes and suffixes.
- Identify root words.
- Improve language understanding.
- Support Natural Language Processing applications.





MORPHEME:



- A **morpheme** is the smallest meaningful unit of language.
- Morphemes play a vital role in word formation and vocabulary expansion.

Example

- Unhappiness

Part	Meaning
Un	Not
Happy	Root word
ness	State or Quality



TYPES OF MORPHEMES

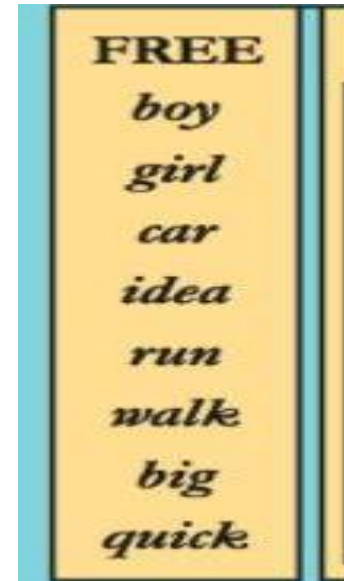


1.Free Morphemes

A Free Morpheme is a morpheme that can stand alone as an independent word and has a complete meaning without needing any additional morphemes.

- **Characteristics**
- Can function as a complete word.
- Carries lexical meaning.
- Can combine with bound morphemes to create new words.

Free Morpheme	Meaning
Book	A Collection Of pages
Run	To move Quickly
Happy	Feeling Joy

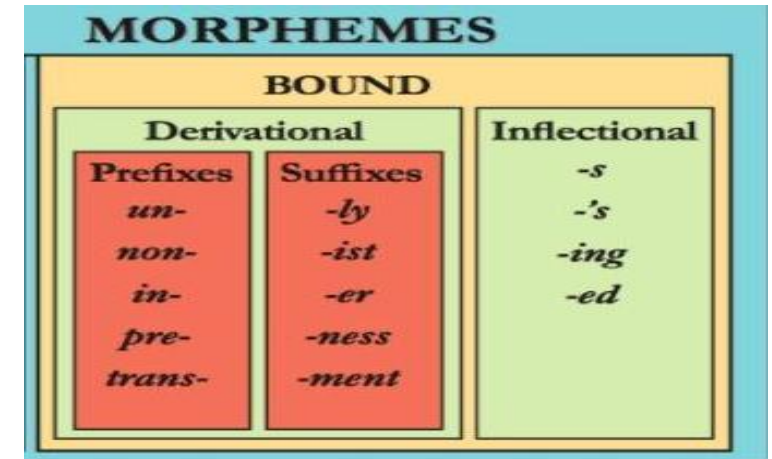




2. Bound Morphemes

A Bound Morpheme is a morpheme that cannot stand alone as a word. It must be attached to a free morpheme to express meaning or grammatical information.

- **Characteristics**
- Cannot exist independently.
- Usually appears as a **prefix** or **suffix**.
- Modifies the meaning or grammatical form of the root word.



Bound Morpheme	Function	Example
Re-	Again	rewrite
-ing	Continuous Action	Playing
-ed	Past Tense	Walked



TYPES OF WORD FORMATION



Process	Description	Example
Prefixation	Adding a prefix before the root word	happy → un happy
Suffixation	Adding a suffix after the root word	teach → teacher
Compounding	Combining two complete words	black + board = blackboard
Blending	Combining parts of two words	breakfast + lunch = brunch
Clipping	Shortening a longer word	examination → exam
Acronyms	Forming a word from initial letters	NASA, UNESCO



IMPLEMENTATION



```
technical presentation example code.py - C:/Users/K Pranitha Reddy/OneD
File Edit Format Run Options Window Help
def analyze_word(word):
    if word == "disagreement":
        print("Word :", word)
        print("Prefix :", "dis")
        print("Root  :", "agree")
        print("Suffix :", "ment")

    elif word == "carelessness":
        print("Word :", word)
        print("Root  :", "care")
        print("Suffix :", "less")
        print("Suffix :", "ness")

    else:
        print("Word not found!")

word = input("Enter a word: ").lower()
analyze_word(word)
```

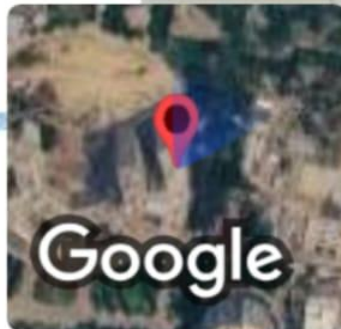
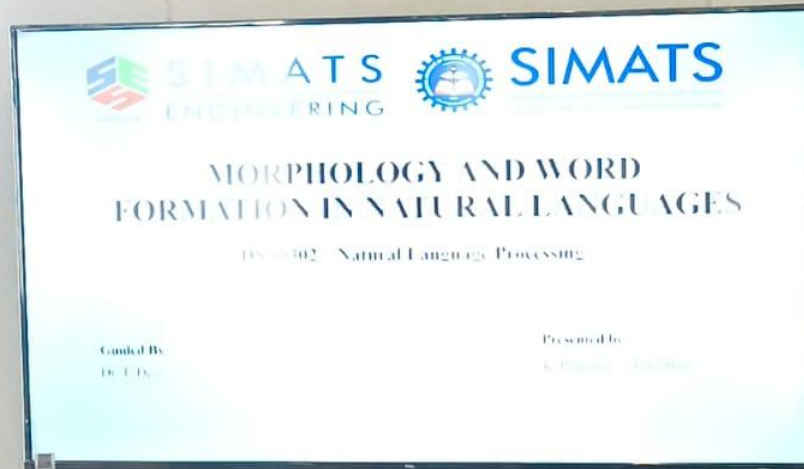
```
IDLE Shell 3.13.7
File Edit Shell Debug Options Window Help
Python 3.13.7 (tags/v3.13.7:bceelc3, Aug 14 2025
AMD64)] on win32
Enter "help" below or click "Help" above for mor
>>>
= RESTART: C:/Users/K Pranitha Reddy/OneDrive/De
sentation example code.py
Enter a word: Disagreement
Word  : disagreement
Prefix : dis
Root   : agree
Suffix : ment
>>> |
```



CONCLUSION



- Morphology studies the structure and formation of words.
- Morphemes are the smallest meaningful units of language.
- Word formation expands the vocabulary of a language.
- Morphology helps NLP systems understand word meanings.
- It improves tasks like translation, spell checking, and text analysis.
- Morphology is a key foundation for modern Natural Language Processing.



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GPS Map Camera

*Thank
you!*